Question Paper

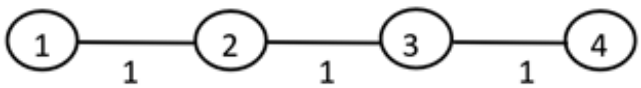
Exam Date & Time: 30-Dec-2024 (02:30 PM - 05:30 PM)

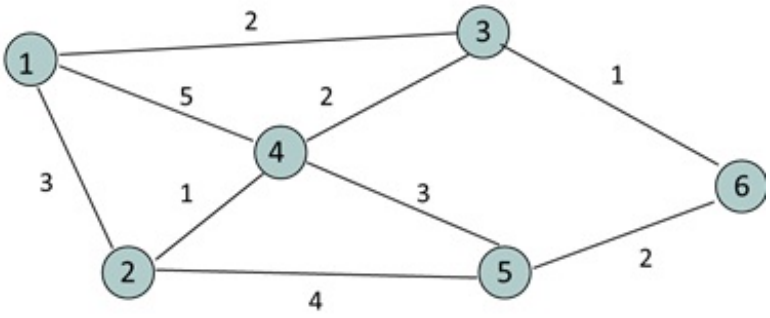
MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 3123/ECE - COMMUNICATION NETWORKS
(MAKEUP)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A)	With a neat diagram, explain the layers of OSI model.	(4)
1B)	Represent the bit stream "11010101" using the following encoding techniques i) Manchester ii) Differential Manchester iii) NRZ-I iv) Polar RZ v) Polar NRZ vi) Unipolar NRZ	(3)
1C)	With a suitable diagram briefly explain the process of message switching. Also mention its advantages and disadvantages.	(3)
2A)	Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).	(4)
2B)	If the packet header is given as: 01 00 F2 03 F4 F5 F6 F7 00 00, find the Checksum.	(3)
2C)	With suitable diagram ,explain Go-Back-N ARQ protocol for both normal operation and lost frame.	(3)
3A)	A network of 'N' stations shares a 128-kbps channel using slotted ALOHA protocol. Each station outputs a 2kb frame on an average of once every 100 seconds, even if the previous one has not yet been sent (e.g., the stations can buffer outgoing frames). What is the maximum possible number of nodes in the network?	(4)
3B)	In a 100Base-FX Ethernet, i) How long does it take for a bit to travel from the beginning to the end of the network? Ignore any propagation delay in the equipment. ii) find the maximum time it takes for a sender to detect a collision. iii) find the minimum size of an Ethernet frame for collision detection iv) Find the bit length in the medium.	(3)
3C)	Consider a 4000m, 16Mbps token ring containing 80 stations. Every station has data to transmit with equal priority. The maximum data allowed to be transmitted is 1000B before giving up the token. Once a station gives up a token how long will it take for that station to get the token again? Assume frame processing time at each station is 4-bits.	(3)
4A)	For the given subnet, update the routing table before and after the link failure between nodes 3 and 4 using Bellman Ford algorithm. 	(4)
4B)	Apply link state routing algorithm for defining the routing table at node '4'.	(3)



4C)	MAHE network having a capacity of 100MB of data arrives at the rate of 25 Mbps for 32sec. The token arrival rate is 5 Mbps and the capacity of bucket is 20MB with a maximum output rate of 25 Mbps. Find: i) burst Length ii) total output time.	(3)
5A)	Explain the connection establishment and release processes in TCP using relevant examples and diagrams.	(4)
5B)	Explain the request and status lines in HTTP.	(3)
5C)	Discuss the steps employed to map the host name to an IP Address using DNS Protocol.	(3)

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